Lightweight Pothole Detection for Vehicle-Based Road Monitoring

Can a small, fast detector find potholes well enough to report them automatically?

CSE36501 Deep Learning — Project Proposal

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November 6, 2026

UNIST, Fall 2026

Motivation

- Today potholes are found by **road-agency inspections** or **citizen reports**.
  - small potholes are rarely reported,
  - potholes on rural / suburban roads are reported late.
- Once autonomous vehicles are common, every car already sees the road $\Rightarrow$ it could detect potholes and report their location automatically.
- But the car's compute is reserved for driving: a monitoring side-task must be efficient, fast, and cheap.

Goal: a pothole detector that is accurate enough *and* light enough to run alongside the driving stack.

Problem statement

Task. Single-class object detection (bounding boxes) of potholes in vehicle-view road images.

Constraint. Low compute: we report accuracy *together with* parameters, GFLOPs and inference speed (FPS).

RQ1. How small and fast can a pothole detector be while keeping useful accuracy? (accuracy–efficiency trade-off)

RQ2. Which efficiency lever costs the least accuracy: a smaller model, a lower input resolution, a lightweight module design, or reduced precision?

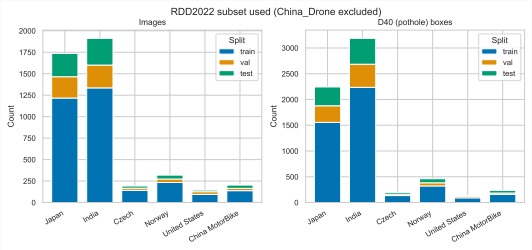
RQ3. Does an efficient model stay reliable in conditions a car will meet — other countries and cameras, and rain?

Datasets

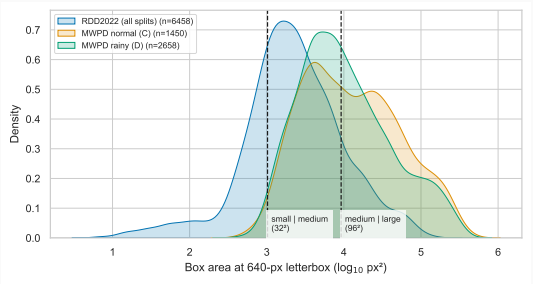
	RDD2022 (Arya et al., 2024)	MWPD (Parvin et al., 2025)
Role	main train / val / test	external robustness test (domain, rain)
Source	vehicle-mounted cameras & phones, 6 countries	mixed web, news and phone images
Size	47,420 images, 4 damage types; we keep **D40 = pothole** (3,610 positive images, 6,458 boxes)	1,357 images, 640×640
Labels, license	Pascal VOC XML; CC BY 4.0	YOLO txt, 1 class; CC BY 4.0
Conditions	clear daylight	normal / rainy; day / twilight / night

- **Why these:** RDD2022 is the largest public multi-country road-damage set, captured from vehicles — the target viewpoint; MWPD adds a different source and real rain.
- China_Drone excluded (top-down drone view $\neq$ vehicle view).
- MWPD ships without weather labels $\Rightarrow$ every image was tagged normal/rainy (pre-tag + manual review).

Dataset analysis



Only 10% of annotated RDD images contain a pothole; Japan + India hold 81% of them.



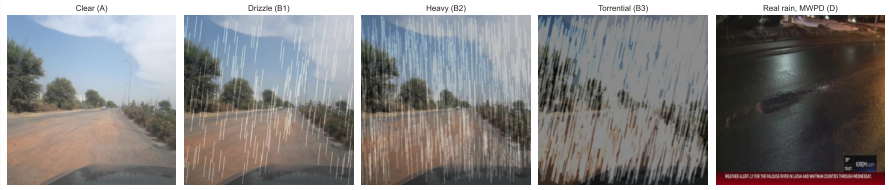
26% of RDD potholes are COCO-small ($< 32^2$ px at 640); MWPD: 1%.

Why it matters: small potholes are exactly the ones people do not report — and the first to disappear when we shrink the model or the input resolution.

Leakage-safe split: blocks of 50 consecutive frames go whole to train/val/test (70/15/15); negatives capped at 20%.

Evaluation design

ID	Composition	Images	Purpose
A	RDD2022 own test split (clear)	708	**main accuracy**
B1/B2/B3	*same images* as A + synthetic drizzle/heavy/torrential	708 each	rain robustness
C	MWPD normal weather	681	new source / camera
D	MWPD rainy	676	real rain



Every model is reported as **accuracy** (mAP50, mAP50-95, recall, small-object AP) **vs. cost** (params, GFLOPs, FPS on GPU and CPU).

B, C, D answer RQ3: does the efficient model still work off the training distribution?

Plan & timeline

Step	Content	Status
1–2	Problem, datasets, splits, evaluation sets A–D	done (this talk)
3	Baselines from tiny to large (YOLOv8n/s, RT-DETR-L as accuracy reference); efficiency levers: input resolution, lightweight own variant, FP16/INT8	next
4	Accuracy–cost curves; small-object AP; robustness on B/C/D; 3 seeds	
5	Analysis, failure cases; report & final video	

Compute: one RTX 3090 (24 GB) for training; speed also measured on CPU as a proxy for in-vehicle hardware. Everything is scripted and seeded for reproducibility.